

Brandon Hesslau

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EDUCATION

Bachelor of Science in Computer Science

DePaul University | Concentration in Real-Time Game Systems

September 2017 – June 2021

Chicago, IL

EXPERIENCE

Senior Software Engineer

August 2024 – May 2025

Independent Project - Desktop Survivors 98

Rochester, MI

- Rapidly prototyped and iterated on novel gameplay mechanics, evolving initial concepts into a shipped commercial title through continuous testing and refinement.
- Architected and shipped a custom C# game engine on MonoGame with high-performance systems for object pooling, entity management, and low-level runtime behavior.
- Designed modular, component-based gameplay architecture emphasizing extensibility and maintainability.
- Shipped a commercial Steam title with Overwhelmingly Positive reviews (95%+), featured by IGN and showcased at PAX 2025, owning the full technical pipeline end-to-end.

Senior Software Engineer

January 2019 – July 2024

Not a Sailor Studios

Chicago, IL

- Collaborated closely with game designers to prototype and implement innovative gameplay mechanics, iterating rapidly based on playtesting feedback.
- Led development of multiple Unity toolchains and editor extensions, reducing iteration time by 40% across engineering, art, and design teams.
- Architected scalable, data-driven gameplay systems and tooling integrated with Unity runtime systems.
- Owned core engine tooling, performance profiling, and cross-platform optimization across PC, Nintendo Switch, PS4, and Xbox while guiding junior engineers.
- Debugged and resolved runtime issues including performance regressions and platform-specific bugs using engine-level profiling and custom diagnostics.

Software Engineer

February 2023 – July 2024

Janus Research Group

Evans, GA

- Developed performance-critical C++/Qt pipeline and workflow tooling converting designer-authored data into runtime-ready Unity components, reducing errors and iteration overhead by 20%.
- Built developer-facing C++ automation and validation tools supporting CI/CD workflows and production tooling maintenance.
- Partnered with engineering, design, and QA to identify pipeline bottlenecks and deliver tools addressing real production pain points.

SELECTED PROJECTS

3D Math Library | C++, SIMD (SSE)

- Header-only math library implementing vectors, matrices, quaternions, and transform hierarchies optimized for real-time rendering and simulation

Aseprite 3D Renderer Plugin | C#, Lua, FBX SDK, Multithreading

- Multithreaded 3D rendering plugin enabling artists to import 3D models and generate configurable 2D sprite sheets for animation workflows

MiniMem - C++ Memory Profiler | C++, CMake, Header-Only Library

- Lightweight, header-only memory profiler for tracking heap allocations and detecting leaks with zero overhead when disabled

SKILLS

Languages: C++, C#, Python, JavaScript, TypeScript, Lua, HLSL/GLSL

Frameworks & Libraries: Unity, MonoGame, Qt, FBX SDK, OpenGL, DirectX

Systems & Engine Development: Memory management, multithreading, data-oriented design, profiling & optimization, custom tooling, build systems (CMake), cross-platform development

Tools: Git, Perforce, Visual Studio, VS Code, CMake, Jira, CI/CD pipelines, Windows / Linux development